LIKHONA ZONYANE

ST10388321

PDDA0801w, GROUP 1

PROGRAMMING FOR DATA ANALYTICS 1 (PDAN8411w)

PART 1

DUE DATE: 31 MARCH 2026

REPORT

Contents

1.INTRODUCTION 3
2.PLANNING OF THE PORTFOLIO OF EVIDENCE 3
3. EXPLORATORY DATA ANALYSIS (EDA) 4
3.1 Data Overview 4
3.2 Data Visualisation 5
4. FEATURE SELECTION AND ENGINEERING 6
5. MODEL DEVELOPMENT 7
6. MODEL EVALUATION 8
7. Coefficient Interpretation 8
8. KEY KINDINGS AND RECOMMENDATIONS 9
9. CONCLUSION 10
REFERENCES 11

1.INTRODUCTION

This report aims to create a linear regression model for a prevalent medical aid scheme in South Africa, predicting medical expenditures based on customer data such as age, gender, BMI, number of children, smoking habits, and geographic location. Müller and Guido (2016) define linear regression, also known as Ordinary Least Squares (OLS), as a supervised learning technique that predicts a continuous target variable based on the linear relationship between input data and output. Linear regression is a popular machine learning approach for estimating variable relations (James et al. 2013).

2.PLANNING OF THE PORTFOLIO OF EVIDENCE

The analysis is classified into five separate steps to ensure a strong and understandable prediction model for the medical aid scheme.

Exploratory Data Analysis (EDA): The first phase entails loading the dataset and thoroughly inspecting for missing values and data types. Summary statistics and distribution plots (histograms and box plots) will be used to assess the skewness of the target variable and the influence of categorical factors such as smoking status.

Feature Selection is a phase where backward elimination will be utilised using P-values to determine the most important predictors. Categorical variables (sex, smoker, region) will be numerically encoded, and features with P-values larger than 0.05 will be deleted to simplify the model and minimise noise.

Model Training: The dataset will be categorised into 80% training and 20% testing sets of data. A Multiple Linear Regression model will be trained using default hyperparameters (including fit_intercept=True) to produce a baseline proof-of-concept.

Interpret and Evaluate Model: The model's performance will be evaluated using R-squared (R^2), Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE). If the initial model contains deficiencies, a retraining step will be carried out, introducing interaction terms (e.g., Smoker * BMI) to reflect nonlinear interactions that occur.

Compile a report: The last stage involves combining the findings into a structured report. This analysis report will discuss the data cleaning process, EDA findings, model training phases, and final suggestions for the medical aid scheme's pricing strategy.

3. EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis (EDA) is a critical phase in data analysis in which we analyse and visualise the data to comprehend its key aspects, identify patterns, and determine how various variables are interconnected (Geeksforgeeks, 2026).

3.1 Data Overview

The dataset contains 1,338 observations across 7 variables. The initial examination validates a clean dataset with no missing values (NaN), assuring useful data integrity during the modelling phase. Each row represents an individual, and each column indicates a characteristic about that individual (Choi, 2018).

- **Age**: A numerical variable denoting the primary beneficiary's age (18-64 years).
- **Sex**: A categorical variable reflecting the beneficiary's gender (male or female).
- **BMI**: Body Mass Index (kg/m²) is a significant indicator of relative body weight.
- Children: A numerical variable indicating the number of dependents covered by the insurance.
- **Smoker**: A categorical variable reflecting smoking status (yes/no).
- **Region**: A categorical variable for residential areas in the United States (four separate regions).
- **Charges**: The target variable represents individual medical expenditures billed.

[16]:
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

[3]:
```
df = pd.read_csv("insurance.csv")
df.head()
```

[3]:

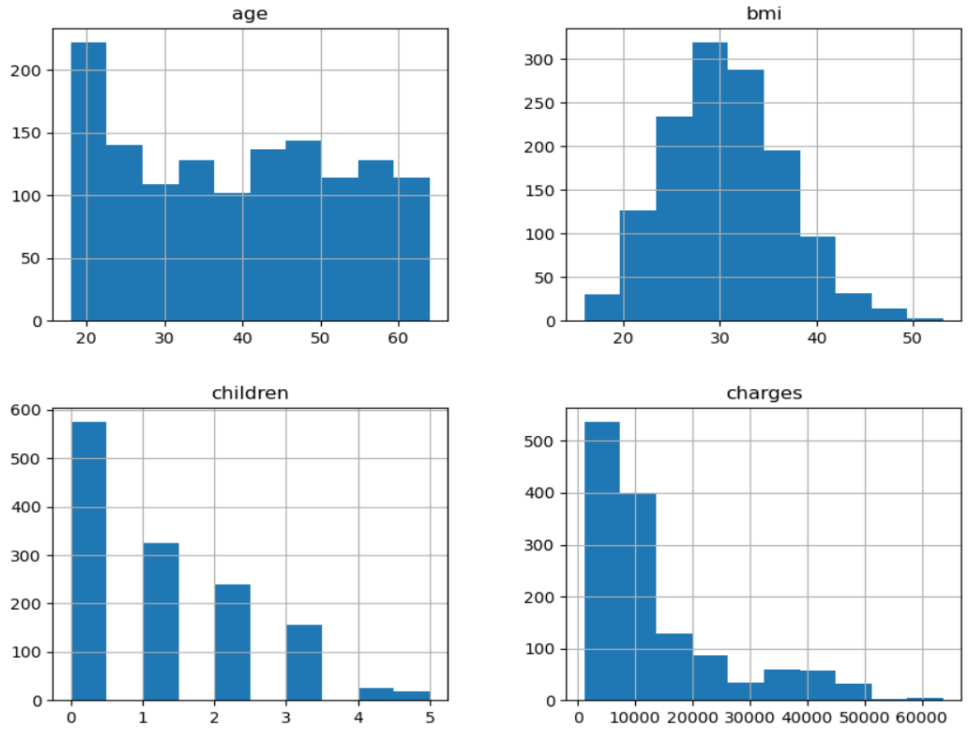
	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

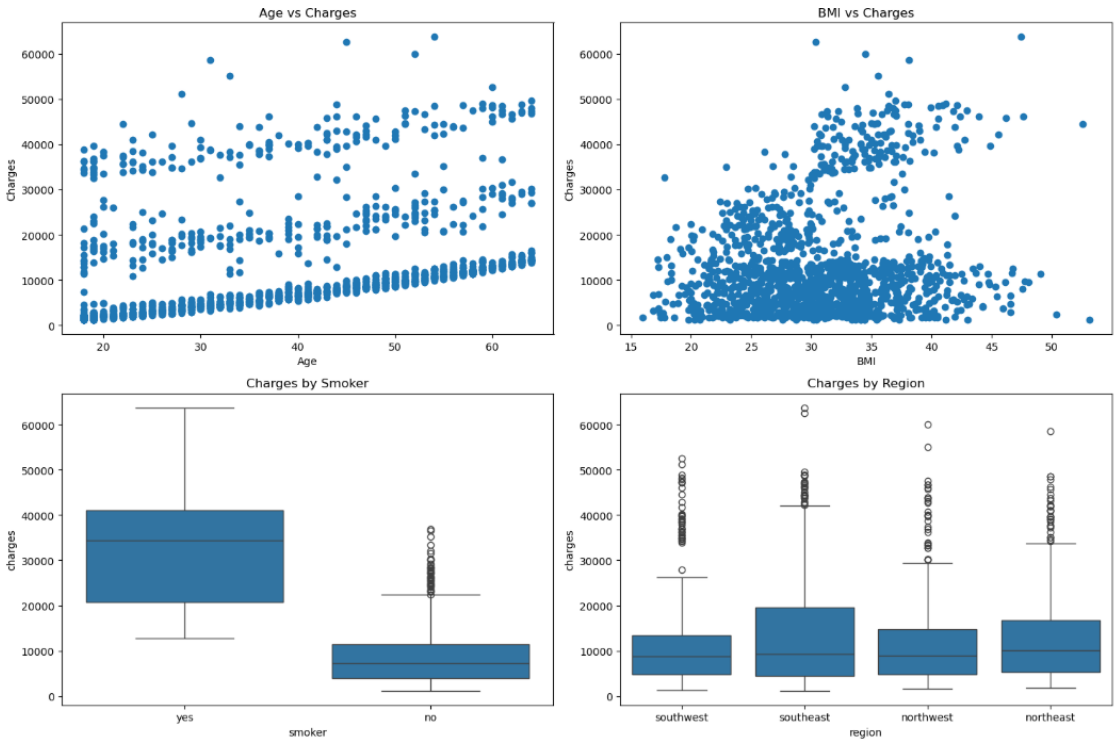
[42]:

	Column	Data Type	Description
0	age	int64	Age of the primary beneficiary
1	sex	object	Gender of the beneficiary (male/female)
2	bmi	float64	Body Mass Index (weight/height²)
3	children	int64	Number of children covered by health insurance
4	smoker	object	Smoker status (yes/no)
5	region	object	Residential area in the US (northeast, southea...
6	charges	float64	Medical charges billed by health insurance

3.2 Data Visualisation

Visualisation techniques such as histograms, scatter plots, and boxplots were utilised to analyse the relationship between variables. The distribution of medical charges was found to be uneven, with some people paying much more than others. There was a considerable correlation between smoking status and medical expenditures, since smokers tend to have much greater expenses. Furthermore, both BMI and age have a positive connection with charges, implying that higher values are often connected with greater prices. Although geographical variances were detected, their influence on charges was not as substantial as other factors.





- Scatter plots show the relationship between the variables.
- Boxplots show the distribution across categories.

4. FEATURE SELECTION AND ENGINEERING

All accessible features were included in the analysis since each one provides important information for forecasting medical expenditures.
The categorical variables (sex, smoker, and region) were converted to numerical representation using one-hot encoding. This phase is required because machine learning models require numerical data.

After encoding, the dataset had attributes like:

- sex_male, smoker_yes, region_northwest, region_southeast, region_southwest

The dataset was then separated into:

Features (X) include all independent variables.
Target variable (y): medical charges.

```
X_train shape: (1070, 8)
X_test shape: (268, 8)
y_train shape: (1070,)
y_test shape: (268,)

X_train sample:
```

	age	bmi	children	sex_male	smoker_yes	region_northwest	region_southeast	region_southwest
560	46	19.95	2	False	False	True	False	False
1285	47	24.32	0	False	False	False	False	False
1142	52	24.86	0	False	False	False	True	False
969	39	34.32	5	False	False	False	True	False
486	54	21.47	3	False	False	True	False	False

```
y_train sample:
560      9193.83850
1285     8534.67180
1142    27117.99378
969      8596.82780
486     12475.35130
Name: charges, dtype: float64
```

5. MODEL DEVELOPMENT

5.1 Selection of Model

Multiple Linear Regression was chosen as the major approach since the target variable (charges) is continuous, and the objective is to comprehend the correlations between variables. This approach is supported by the need for a "sliding scale" of charges, as linear regression produces unambiguous, interpretable coefficients that explicitly show how each feature affects the ultimate cost (Müller & Guido, 2017).

5.2 Model Training and Hyperparameter Tuning

To ensure repeatability, the dataset was divided into a training set (80%, 1,070 samples) and a testing set (20%, 268 samples) using a set random_state. The Linear Regression model was trained utilising the training data. Standard options were selected since Linear Regression does not require considerable hyperparameter modification to build a baseline.

6. MODEL EVALUATION

The model was assessed using the testing dataset and the following metrics:
Mean squared error (MSE): 33,596,915.85
R^2 score: 0.7836.

The MSE measures the average squared difference between expected and actual data. The model explains 78.36% of the variance in medical expenses, as indicated by the R^2 value.
These findings show that the model performs well and can capture the association between characteristics and medical charges.

Model Evaluation

MSE measures average error in predictions. R^2 indicates how well the model explains variance in charges. Higher R^2 → better model.

[56]:
```
from sklearn.metrics import mean_squared_error, r2_score

mse = mean_squared_error(y_test, predictions)
r2 = r2_score(y_test, predictions)

print("Mean Squared Error (MSE):", mse)
print("R-squared (R²):", r2)
```

```
Mean Squared Error (MSE): 33596915.85136146
R-squared (R²): 0.7835929767120723
```

7. Coefficient Interpretation

Feature coefficients demonstrate how each predictor influences the projected medical charges:

Interpretation of Coefficients

Smoking has the largest positive coefficient → biggest increase in costs. BMI and age also increase charges. Useful for understanding which factors most influence medical costs.

[57]:
```
coefficients = pd.DataFrame({
    'Feature': X.columns,
    'Coefficient': model.coef_
})
coefficients
```

[57]:

	Feature	Coefficient
0	age	256.975706
1	bmi	337.092552
2	children	425.278784
3	sex_male	-18.591692
4	smoker_yes	23651.128856
5	region_northwest	-370.677326
6	region_southeast	-657.864297
7	region_southwest	-809.799354

The smoker_yes coefficient (23,651.13) indicates that smoking significantly raises projected medical expenditures, while other factors remain unchanged.
BMI and age can have a favourable impact on costs.
Negative coefficients for the area variables indicate small declines in charges compared to the baseline region.

These data confirm that smoking and body composition are significant cost factors.

8. KEY KINDINGS AND RECOMMENDATIONS

The analysis provided numerous crucial findings, which may be immediately applied to the medical aid scheme's pricing approach.

- Finding 1: The number of children has a substantial influence, but gender and area have very little effects on ultimate charges.
- Finding 2: Smoking has the greatest influence on medical bills, with a very strong positive coefficient that adds an estimated $23,850 to yearly costs.
- Finding 3: BMI and age have a considerable impact on medical expenditures, with each year of age raising the premium by almost $257.

Recommendations:

- Recommendation 1: Premium Adjustments - When calculating rates, insurance organisations should prioritise smoking status and apply a considerable "smoker's surcharge" to represent the real risk.
- Recommendation 2: Model Refinement - Including more factors, such as medical history or lifestyle behaviours, may enhance forecast accuracy in subsequent iterations.
- Recommendation 3: Health Program - Health programs that promote smoking cessation and weight management might dramatically cut medical expenditures for both the plan and its members.

9. CONCLUSION

In conclusion, the Linear Regression model accurately predicted medical insurance charges for the evaluated medical aid scheme. The model shows that smoking status, BMI, and age are the most important factors influencing expenses, whereas gender and geographic location have little influence.

The strong R^2 value (0.78) suggests that the model explains a significant amount of the variability in medical expenditures, validating its dependability for decision-making. These insights can help medical aid scheme establish fair and risk-adjusted rates, create targeted health initiatives, and optimise resource allocation.

Prospective improvements might involve adding new variables, such as medical history, lifestyle characteristics, or chronic illnesses, to increase predicted accuracy and provide a more complete pricing approach.

REFERENCES

Choi, M. (2018). Medical Cost Personal Datasets. Available at: https://www.kaggle.com/datasets/mirichoi0218/insurance [Accessed: 20 March 2026].

Geeksforgeeks. (2026). What is Exploratory Data Analysis? Available at: https://www.geeksforgeeks.org/data-analysis/what-is-exploratory-data-analysis/ [Accessed 26 March 2026].

Müller, A.C. & Guido, S. (2017). *Introduction to Machine Learning with Python.* O'Reilly Media.

Scikit-learn Documentation. (2024). Linear Regression. Available at: https://scikit-learn.org/ [Accessed: 26 March 2026].

The Independent Institute of Education. (2024). Programming for Data Analytics 1 (PDAN8411) Module Manual.